

## Agile vs Waterfall - Notes

### 1. Introduction

Software development methodologies define how projects are planned, developed, tested, and delivered.

The two major approaches are:

- Waterfall Model: Traditional sequential development approach.
- Agile Model: Flexible iterative development approach.

### WATERFALL MODEL

Definition:

Waterfall is a linear sequential software development model where each phase is completed before moving to the next phase.

Phases:

#### 1. Requirement Analysis

- Collect and document customer requirements.

#### 2. System Design

- Create architecture and design documents.

#### 3. Implementation

- Developers write code according to the design.

#### 4. Testing

- Complete system testing is performed.

#### 5. Deployment

- Software is released to users.

#### 6. Maintenance

- Bug fixes and updates are provided.

Advantages:

- Simple and easy to understand.
- Clear project structure.
- Easy project management.
- Suitable for fixed requirements.
- Good documentation.

Disadvantages:

- Difficult to change requirements.
- Customer feedback comes late.
- Testing happens after development.
- Higher risk for unclear requirements.
- Longer delivery time.

When to use Waterfall:

- Requirements are fixed.
- Projects need strict documentation.
- Government or safety-critical projects.

## AGILE MODEL

Definition:

Agile is an iterative and incremental approach where software is developed in small cycles called sprints.

Agile Process:

### 1. Planning

- Requirements are prioritized.

### 2. Sprint Development

- Features are developed in short cycles.

### 3. Testing

- Testing is performed continuously.

#### 4. Review

- Customer reviews completed features.

#### 5. Retrospective

- Team discusses improvements.

#### Advantages:

- Flexible to changing requirements.
- Faster delivery.
- Continuous customer feedback.
- Early problem detection.
- Better collaboration.

#### Disadvantages:

- Requires experienced teams.
- Final cost and timeline are harder to predict.
- Less documentation.
- Requires customer involvement.

#### When to use Agile:

- Requirements change frequently.
- Customer feedback is important.
- Complex software projects.

### DIFFERENCE BETWEEN AGILE AND WATERFALL

#### Waterfall:

- Sequential approach.
- Requirements fixed at beginning.
- Testing after development.

- Single final release.
- Heavy documentation.
- Low customer involvement.

Agile:

- Iterative approach.
- Requirements can change.
- Continuous testing.
- Frequent releases.
- Less documentation.
- High customer involvement.

## EXAMPLES

Waterfall Example:

Banking systems with fixed regulations and requirements.

Agile Example:

Mobile applications where features change based on user feedback.

## KEY POINTS

- Waterfall = Plan everything first -> Build -> Test -> Deliver.
- Agile = Plan -> Develop -> Test -> Feedback -> Improve.
- Waterfall focuses on process and documentation.
- Agile focuses on flexibility and customer satisfaction.